Phys 101 Worksheet

Comparing Model and Experiment

Name: _____ **Student #:** _____ **Student Email #:** _____

1. **Thinking Like a Physicist**
   Suppose there's a theory "Dogs cannot be born with naturally blue fur." Can this theory be proven to be true? **Explain** why or why not.

2. **Dimensional Analysis**
   Suppose you're a butterfly researcher. You find that rarer butterflies appear to be faster, and flutter their wings at a higher rate than more common butterflies. You invent a new unit, a "Whimsy" (denoted as W), which is the **product** the maximum velocity of a given species of butterfly, v, and the frequency at which it flaps its wings f.

   (a) Explicitly write out the formula to calculate Whimsiness. What are it's dimensions?

   (b) Suppose now you want to create a formula that could predict the most whimsical butterfly any given butterfly catcher could catch knowing only their mass, m and the maximum force the can swing with F_s. Write a possible solution (within a constant) that could be a solution to the problem.

   (c) Does the formula you've devised make sense? Explain wy or why not. (Hint: When you increase certain variables, does whimsiness increase or decrease as you would expect?)